

Tilak Patel

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EDUCATION

Northeastern University

Boston, MA

B.S. Computer Science | GPA: 3.6/4.0

May 2027

Coursework: Grad AI, Grad Generative AI, Grad HPC, Algorithms, Computer Systems, Systems Security, Robotics

EXPERIENCE

Bose Corporation

Framingham, MA

Software Engineer Co-op

Jan 2026 – June 2026

- Built full-stack log analysis platform with 60+ REST endpoints, streaming, and session persistence using FastAPI
- Designed 3-phase processing pipeline with 25+ custom tools, reducing firmware debugging from hours to minutes
- Automated defect triage via CI pipeline that processes Jira tickets and posts reports without human input
- Shipped cross-platform desktop app (React, Electron, Copilot SDK) integrating Jira, Confluence, and Artifactory

Empowerreg.ai

Remote

AI Engineer Intern (Part-Time)

July 2025 – Dec 2025

- Built interactive visualization tool mapping FDA complaint severity with heatmaps for medical device risk analysis
- Deployed Grafana and Loki observability stack, unifying real-time log monitoring across 5+ microservices
- Saved 4+ hours/week per analyst by building AI-powered assistant for regulatory compliance queries

Pendar Technologies

Boston, MA

Laser Software Engineer Co-op

July 2025 – Dec 2025

- Built real-time data acquisition app in Qt/PySide6 for laser sensor testing, fully replacing legacy LabVIEW system
- Implemented automated overnight testing with CSV export, eliminating hours of manual data collection daily

SRC, Inc.

Syracuse, NY

Machine Learning Engineer Intern

April 2025 – July 2025

- Built pipeline extracting knowledge triplets from 500+ radar docs at 90% accuracy using Pydantic and Docling
- Fed extracted triplets into knowledge graphs via LangGraph, automating analysis and cutting review time by 70%

PROJECTS/RESEARCH

Research – GPU Checkpoint-Restart (Prof. Gene Cooperman) | C, MPI, NCCL, Linux

- Profiled MPI checkpoint-restart overhead in MANA using flamegraphs, identifying communication bottlenecks
- Implementing GPU checkpoint-restart for NCCL collectives, enabling fault recovery in multi-GPU training jobs

Game Boy Emulator | C++, SDL3, CMake

[GitHub](#)

- Developed Game Boy (DMG) emulator passing Blargg's CPU test suite, implementing all 500+ LR35902 opcodes
- Implemented scanline-based PPU and memory-mapped I/O via SDL3, running Game Boy ROMs at 60 FPS

Devspace – HackBeanpot 1st Place | Java, React, Spring Boot, Docker, CUDA, WebSockets

[GitHub](#)

- Built collaborative cloud IDE with GPU-accelerated Docker execution (CUDA/C++/Python) on Jetson Nano
- Implemented real-time multi-user editing with cursor sync and conflict resolution via WebSockets

Open Source – [github/awesome-copilot](#) (29K+ stars) | Python, Copilot SDK

[GitHub](#)

- Wrote error-recovery hooks and PyInstaller frozen-build recipes for the Copilot SDK, merged into official repo

TECHNICAL SKILLS

Languages: Python, C/C++, Java, C#/.NET, SQL, Assembly (x86)

Frameworks: FastAPI, Spring Boot, React, Node.js, Electron, PyTorch, NumPy, LangGraph, Qt/PySide6

Infrastructure: Docker, Git, Linux, AWS, GitHub Actions, CUDA, NCCL, MongoDB, Grafana, Jira/Confluence API